

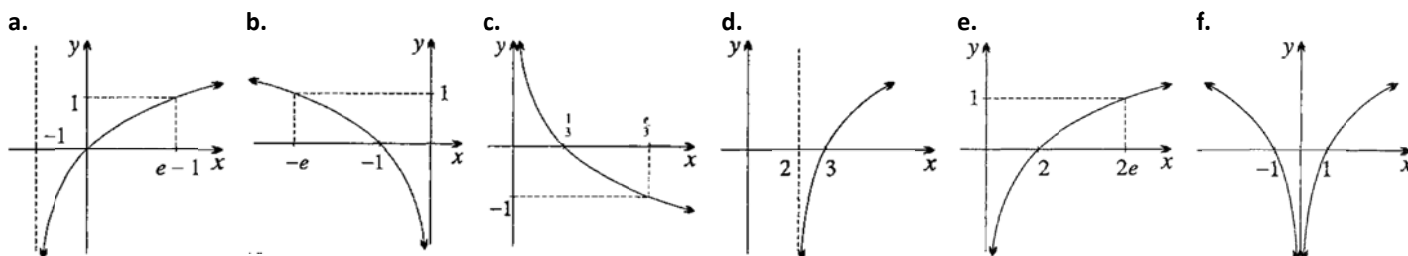
Logarithm function – Applications of Differentiation

1. Use your knowledge of transformations to help sketch the graphs of the given functions:
 - a. $y = \ln(x + 1)$
 - b. $y = \ln(-x)$
 - c. $y = -\ln 3x$
 - d. $y = \ln(x - 2)$
 - e. $y = \ln\left(\frac{\pi}{2}\right)$
 - f. $y = \ln|x|$
2. a. Write down the domain of $y = \ln(1 + x^2)$
 - b. Show that $y' = \frac{2x}{1+x^2}$ and $y'' = \frac{2(1-x^2)}{(1+x^2)^2}$.
 - c. Hence show that $y = \ln(1 + x^2)$ has one stationary point, and determine its nature.
 - d. Find the coordinates of the two points of inflexion.
 - e. Hence sketch the curve, and then write down its range.
 - f. Hence sketch $y = \ln\left(\frac{1+x^2}{2}\right)$. [Hint: You will need to use the logarithm laws.]
3. a. Find the domain of $y = (\ln x)^2$.
 - b. Find y' and y'' , and hence show that the curve has as an inflexion at $x = e$.
 - c. Classify the stationary point at $x = 1$, sketch the curve and write down the range.
4. a. Determine the first two derivatives of $y = x - \ln x$
 - b. Deduce that the curve is concave up for all values of x in its natural domain.
 - c. Find the minimum turning point.
 - d. Sketch the curve and write down its range.
5. a. Write down the domain of $y = \frac{1}{x} + \ln x$.
 - b. Find the first and second derivatives.
 - c. Show that the curve has a minimum at $(1, 1)$ and an inflexion at $(2, \frac{1}{2} + \ln 2)$.
 - d. Sketch the graph and write down its range.
6. a. Write down the domain of $y = \frac{\ln x}{x}$, then find any horizontal or vertical asymptotes.
 - b. Find y' and y''
 - c. Find any stationary points and determine their nature.
 - d. Find the exact coordinates of the lone point of inflexion.
 - e. Sketch the curve and write down its range.
7. a. Write down the equation of the tangent to $y = 2 \ln x$ at the point where $x = c$. Hence find any values of c for which the tangent passes through the origin.
 - b. Repeat part (a) for these curves:
 - i. $y = (\ln x)^2$
 - ii. $y = x^2 \ln x$
8. a. Show that the equation of the tangent to $y = x \ln x$ at the point $x = e$ is $y = 2x - e$
 - b. Find the distance from this tangent to the origin.

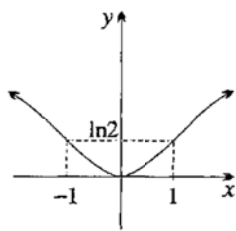
9. a. Find the equation of the tangent to $y = (\ln x)^2$ at the point where $x = t$.
 b. Find the area of the triangle cut off by this tangent and the coordinate axes when $t = e$.
10. Investigate the curve $y = -x \ln x$ as follows, then sketch it and write down its range.
 a. Find its domain and any intercepts.
 b. Find and classify any stationary points.
 c. Examine the behaviour of y and y' as $x \rightarrow 0^+$
11. a. Find and classify the lone stationary point of $y = x^2 \ln x$ in its natural domain.
 b. Show that there is an inflexion at $x = e^{-\frac{3}{2}}$.
 c. Examine the behaviour of y and y' as $x \rightarrow 0^+$.
 d. Hence sketch the graph of this function, then write down its range
12. Carefully classify the critical points of $y = \frac{x}{\ln x}$ and show that there is an inflexion at $(e^2, \frac{1}{2}e^2)$. Examine the behaviour of y and y' as $x \rightarrow 0^+$ and as $x \rightarrow \infty$, then sketch the curve and write down its range.
13. a. Write down the domain of $y = \ln\left(\frac{x^2}{x+1}\right)$
 b. Show that $y' = \frac{x+2}{x(x+1)}$
 c. Show that $y' = 0$ at $x = -2$. Explain why there is no stationary point there.
 d. How many inflexion points does this curve have?
 e. Sketch the graph.
14. a. What is the natural domain of $y = \ln(\ln x)$?
 b. What is the x -intercept?
 c. Find y' and y'' , and explain why there is no stationary points.
 d. Confirm that $y''\left(\frac{1}{e}\right) = 0$, and explain why there is no inflexion there.
 e. Sketch the curve.
15. Two engines, Thomas and Henry, move on close parallel tracks. They start at the origin, and are together again at time $t = e - 1$. Thomas' displacement-time equation, in units of metres and minutes, is $x = 300 \ln(t + 1)$, and Henry's is $x = kt$, for some constant k .
 a. Sketch the two graphs.
 b. Show that $k = \frac{30}{e-1}$.
 c. Use calculus to find the maximum distance between Henry and Thomas during the first $e - 1$ minutes, and the time when it occurs (in exact form, and then correct to the nearest metre or the nearest second).

Answer

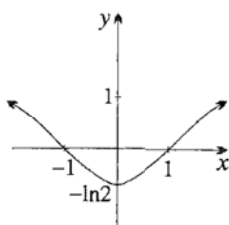
1.



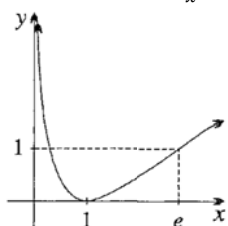
2. **a.** All real x
e. $y \geq 0$



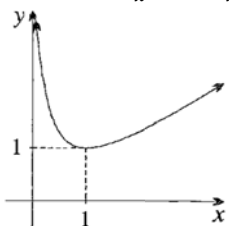
- f.** all real x



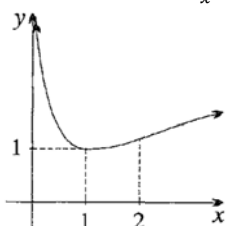
3. **a.** $x > 0$; **b.** $y' = \frac{2}{x} \ln x$, $y'' = \frac{2}{x^2} (1 - \ln x)$; **c.** $y \geq 0$



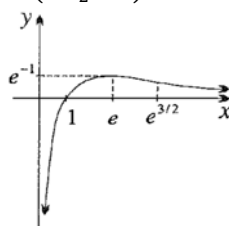
4. **a.** $y' = 1 - \frac{1}{x}$, $y'' = \frac{1}{x^2}$; **b.** $x > 0$; **d.** $y \geq 1$



5. **a.** $x > 0$; **b.** $y' = \frac{x-1}{x^2}$ and $y'' = \frac{2-x}{x^3}$; **d.** $y \geq 1$

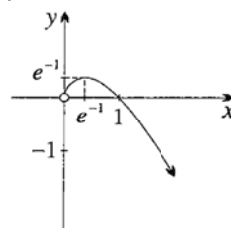


6. **a.** $x > 0$. $y \rightarrow 0^+$ as $x \rightarrow \infty$ so the x -axis is a horizontal asymptote, $y \rightarrow \infty$ as $x \rightarrow 0^+$ so the y -axis is a vertical asymptote; **b.** $y' = \frac{1}{x^2} (1 - \ln x)$ and $y'' = \frac{1}{x^3} (2 \ln x - 3)$; **d.** $(e^{\frac{3}{2}}, \frac{3}{2} e^{-\frac{3}{2}})$; **e.** $y \leq e^{-1}$

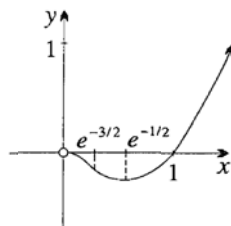


7. **a.** $y - 2 \ln c = \frac{2}{c} (x - c)$, $c = e$; **b. i.** $c = 1$ or e^2 , **ii.** e^{-1}
8. **b.** $\frac{e}{\sqrt{5}}$
9. **a.** $y = \frac{2 \ln t}{t} x - 2 \ln t + (\ln t)^2$; **b.** $\frac{e}{4}$

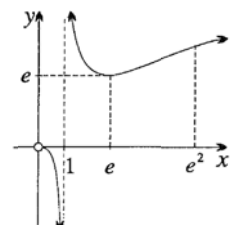
10. **a.** $x > 0$; **b.** $y \rightarrow 0^+$ as $x \rightarrow 0^+$, $y' \rightarrow \infty$ as $x \rightarrow 0^+$, hence the graph becomes vertical approaching the origin.
 $y \leq e^{-1}$.



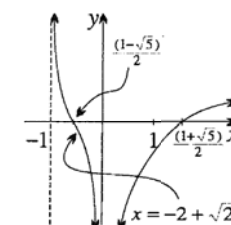
11. **a.** $x > 0$. Minimum at $(\frac{1}{\sqrt{e}}, -\frac{1}{2e})$; **c.** $y \rightarrow 0$ as $x \rightarrow 0^+$, $y' \rightarrow 0$ as $x \rightarrow 0^+$, hence the graph becomes horizontal approaching the origin; **d.** $y \geq -\frac{1}{2e}$



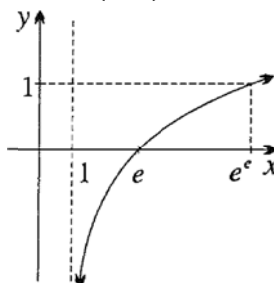
12. $x > 0$, $x \neq 1$, $y < 0$ or $y \geq e$. $x = 1$ is a vertical asymptote and the curve becomes horizontal approaching the origin.



13. **a.** $x > -1$, $x \neq 0$; **c.** $x = -2$ is outside the domain; **d.** one at $x = -2 + \sqrt{2}$



14. **a.** $x > 1$; **c.** $y' = \frac{1}{x \ln x}$, which can never be zero,
 $y'' = -\frac{1 + \ln x}{(x \ln x)^2}$; **d.** the value is outside the domain.



15. **a.** see below; **c.** The maximum distance is $300 \ln(e - 1) - \frac{300(e-2)}{e-1} \div 37$ metres when $t = e - 2 \div 43''$.

